MIL-W-51114A(EA)
10 November 1976
SUPERSEDING
MIL-W-51114(MU)
4 March 1966

MILITARY SPECIFICATION

WATER TESTING KIT, CHEMICAL AGENTS, AN-M2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a water testing kit for chemical agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-C-265 - Chemicals, Analytical; General Specification For.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-P-50039 - Paper, Nitrazine.

FSC 6665

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Edgewood Arsenal, Attn: SAREA-TS-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

Fed-Std-356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

EDGEWOOD ARSENAL

D5-77-49 - Water Testing Kit, Chemical Agents, AN-M2.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION

The National Formulary

(Copies of the National Formulary may be obtained upon application to the J. B. Lippincott Company, Philadelphia, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D5-77-49 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings. Unless otherwise specified, all chemicals shall conform to Specification O-C-265 and be no more than one year old at the time of assembly.

3.1.2 Components. All components of the water testing kit shall conform to the specifications and subsidiary drawings listed on Drawing D5-77-49.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the currentversion before use.

3.1.2.1 Reagent A. Reagent A shall be in tablet form, white in color, and shall consist of 99 parts potassium bisulfate (arsenic content shall not exceed 0.0001 percent), 1 part copper sulfate pentahydrate, and the necessary binder and lubricant. Each tablet shall be approximately 13/32 inch in diameter and 5/32 inch thick and be wrapped in approximately 1 inch by 1 7/16 inches by 0.35 mil thick aluminum foil wax-mounted to tissue. The tissue side shall be in contact with the tablet. The aluminum foil shall be the same color as the cap for reagent A vial. Two tablets shall dissolve without foaming in 10 milliliters (ml) of water at 25°C in 10 minutes and shall produce no stain on the arsenic test paper in the absence of arsenic when tested as specified in 4.5.1.1 (blank).

3.1.2.2 Reagent B. Reagent B shall be arsenic test papers (see 6.3) 1/16 inch wide (tolerance + 1/32 inch, -0 inch) by 2 1/2 inches long (tolerance ± 1/16 inch). Arsenic test papers shall give a 1/4 inch stain within 20 minutes when tested as specified in 4.5.1.1.

3.1.2.3 Reagent C. Reagent C shall be zinc pellets; each pellet shall weigh 0.4 ± 0.2 gram (g).

3.1.2.4 Reagent D. Reagent D shall conform to the composition shown in table 1. Each tablet shall be approximately 7/32 inch in diameter, flat or convex, and shall contain no more than 0.1037 g or less than 0.0972 g of the mixture. Magnesium stearate may be added in an amount not to exceed 0.25 percent of the total weight. No mineral oil may be added. The tablet shall dissolve in 1 ml of water at 25°C in 3 minutes when tested as specified in 4.5.1.2 (blank). Reagent D with reagent E (3.1.2.5) shall give a positive test for mustard when tested as specified in 4.5.1.2.

Table 1. Composition of reagent D

Ingredient	Specification	Percent by weight
DB-3 nitrate		3.00
Potassium perchlorate	O-C-265 (USP)	5.00
Urea	O-C-265 (NF)	92.00

3.1.2.5 Reagent E. Reagent E shall be in tablet form and shall consist of 0.250 g of anhydrous potassium carbonate (ACS), a maximum of 0.0162 g of magnesium stearate, and talcum powder. No mineral oil may be added. Each tablet shall be approximately 1/4 inch in diameter, flat or convex. Each tablet shall be wrapped in approximately 1 inch by 1 7/16 inches of 0.35 mil aluminum foil wax-mounted to tissue. The

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

tissue side shall be in contact with the tablet. The aluminum foil shall be the same color as the cap for reagent E vial. Two tablets shall dissolve in water at 25°C in 3 minutes when tested as specified in 4.5.1.2 (blank). Reagent E with reagent D (3.1.2.4) shall give a positive test for mustard when tested as specified in 4.5.1.2.

3.1.2.6 Reagent F. Reagent F shall be pH test paper conforming to the requirements of Specification MIL-P-50039 and shall pass the test specified in 4.5.1.3. The approximate size of each paper shall be 2 1/8 inches by 3/16 inch.

3.1.2.7 Reagent G. Reagent G shall be halazone tablets conforming to the requirements of the National Formulary. The tablets, before and after aging for two weeks, shall contain no less than 1.8 milligrams (mg) or more than 2.5 mg of available chlorine per tablet when tested as specified in 4.5.1.4.

3.1.2.8 Reagent H. Reagent H shall be in tablet form and shall conform to the composition shown in table II. Each tablet shall weigh 0.042 ± 0.004 g and shall dissolve without foaming in 25 ml of water at 25°C in 3 minutes when tested as specified in 4.5.1.5 (blank). Reagent H with reagent I (3.1.2.9), reagent J (3.1.2.10), and reagent K (3.1.2.11) shall give a positive test for GB when tested as specified in 4.5.1.5.

Table II. Composition of reagent H

Ingredient	Specification	Weight in grams
Polyethylene glycol 6000	See 6.5	0.005
Acacia, powdered	O-C-265 (USP)	0.0001
Lactose	O-C-265 (USP)	0.032
Ortho-dianisidine dihydrochloride	See 6.5	0.005 ± 0.0002

3.1.2.9 Reagent I. Reagent I shall be in tablet form and shall conform to the composition shown in table III. Each tablet shall weigh 0.295 ± 0.030 g, shall be approximately 1/4 inch in diameter, and shall be wrapped in approximately 1 inch by 1 7/16 inches by 0.35 mil thick aluminum foil was mounted to tissue. The tissue side shall be in contact with the tablet. The aluminum foil shall be the same color as the cap for reagent I vial. Two tablets shall disintegrate without foaming in water at 25°C in 7 minutes when tested as specified in 4.5.1.5 (blank). The pH of the solution (blank) shall be no less than 7.7. Reagent 1 with reagent H (3.1.2.8), reagent J (3.1.2.10), and reagent K (3.1.2.11) shall give a positive test for GB when tested as specified in 4.5.1.5.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

Table III. Composition of reagent I

Ingredient	Specification	Weight in grams
Potassium carbonate	O-C-265 (ACS)	0.005 ± 0.001
Sodium chloride	O-C-265 (ACS)	0.098
Magnesium stearate	O-C-265 (ACS)	0.0015
Sodium hexametaphosphate	See 6.6	0.190 ± 0.01

3.1.2.10 Reagent J. Reagent J shall consist of 5 g of Du Pont "Perdox" powder. It shall have a minimum active oxygen content of 14.5 percent when tested as specified in 4.5.1.6, and 13.5 percent after an additional 2 weeks (total of 4 weeks) storage at 65°C. Reagent J with reagent H (3.1.2.8), reagent I (3.1.2.9), and reagent K (3.1.2.11) shall give a positive test for GB when tested as specified in 4.5.1.5.

3.1.2.11 Reagent K. Reagent K shall be a colorless mixture of 25 ± 5 parts by volume of acetone (see 6.7) and 75 ± 5 parts by volume of xylene in ampules. Each ampule shall contain 1.0 ± 0.1 ml of reagent K when tested as specified in 4.5.2 and shall not leak when tested as specified in 4.5.3.

3.1.2.12 Reagent X. Reagent X shall be in tablet form; each tablet consisting of no less than 0.00080 g orthotolidine dihydrochloride, 0.0850 g potassium bisulfate (USP), and sufficient sodium chloride to bring the total weight of the tablet to 0.22 ± 0.03 g. The orthotolidine dihydrochloride shall be of a good commercial quality. One tablet shall indicate the residual chlorine, if present, in water when tested as specified in 4.5.1.7 (see 6.9).

3.1.2.13 Test vial for X color band. The color of the band on the test vial X shall match the yellow color produced by 1.0 ± 0.2 ppm residual chlorine when tested as specified in 4.5.4.

3.1.2.14 Bottle P color band. The color of the band on bottle P shall match the yellow color produced by a 0.0012 ± 0.0001 M potassium dichromate solution when tested as specified in 4.5.5.

3.1.2.15 Instructions. The instruction booklet shall be manufactured and assembled in accordance with the appendix to this specification.

3.2 Manufacture and assembly. The kit shall be manufactured and assembled as specified on Drawing D5-77-49.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

3.3 Preproduction sample. When specified in the contract or purchase order, a preproduction sample of water testing kits shall be produced in accordance with this specification for examination and test.

3.4 Workmanship. The kit and components shall be free from damage (cracks, punctures, tears, chipped corners, and broken tablets or vials) and contamination (dirt, grease, and oil).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Government's responsibility. The Government will be responsible for the performance of the tests specified in 4.5.1.2 and 4.5.1.5. Sample tablets, applicable to the tests, shall be forwarded to the laboratory designated by the contracting officer.

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

4.3 Preproduction inspection.

4.3.1 Preproduction sample. Prior to the start of regular production, a preproduction sample of water testing kits of the number specified in the contract or order, shall be produced using the same methods, materials and equipment as will be used in regular production.

4.3.2 Examination and test. Sample water testing kits shall be examined and tested for all the requirements of this specification.

4.3.3 Acceptance/rejection. The preproduction sample shall comply with all the requirements of this specification to be acceptable.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the water testing kits offered for acceptance which have been produced by one manufacturer under essentially the same manufacturing conditions. No more than one lot of chemical components shall be represented in one lot of kits.

4.4.2 Sampling.

4.4.2.1 For nondestructive examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Sampling shall be conducted in accordance with level S-1 of MIL-STD-105.

4.4.3 Inspection provisions.

4.4.3.1 For examination. Sample water testing kits and level A preparation for delivery thereof shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For test. Sample kits shall be tested as specified in 4.5. The acceptance number is zero.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects.

(a) Water testing kit, chemical agent, AN-M2, assembly (Drawing D5-77-49).

Categories	Defects	Acceptance standards
*1	Ampule leakage	4.5.3
Major:	AQL 1.0 percent defective	
101	Component missing, incorrect, or incorrectly assembled	
102	Instruction booklet incorrect, illegible, or missing	
103	Component marking or color coding incorrect, missing, or illegible	
104	Cotton padding in compartments missing	
105	Tape missing or incorrect	
106	Cushioning material in lid of the case missing or incorrect	
107	Damage (see 3.4)	
108	Tablets not wrapped (when required)	
109	Wrapping foil color incorrect (when required)	
Minor:	AQL 4.0 percent defective	
201	Contamination (see 3.4)	

*Each ampul in the lot shall be examined for this defect prior to assembling the kit.

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.5 percent defective
101	Unit or shipping container not as specified
102	Cushioning not as specified
103	Closure of unit or shipping container not as specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

104	Number of kits per shipping container not as specified
105	Unit or shipping container damaged
106	Marking of unit or shipping container illegible, incorrect, or incomplete

4.5 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Prior to chemically testing the kit, reagent X and one half of the reagent G tablets shall be removed and the remainder of the kit aged for 2 weeks at 65° ± 2°C. After aging, the kit shall be tested as follows:

4.5.1 Reagents.

4.5.1.1 Reagents A and B. Prepare an aqueous solution containing 5 ppm arsenic. Fill bottle P to the 25 ml mark with the arsenic solution; add 2 tablets from vial A and swirl to dissolve. Carefully insert a paper test strip from vial B into tube L, bending the paper test strip near the top so that it will remain in an upright position (only the top end of the paper test strip should be touched to insure the paper test strip being dry). When the tablets from vial A have dissolved in bottle P, add 5 pellets from vial C. Immediately fit the tube L assembly into bottle P and let react for 20 minutes. If the solution is cold, warm the contents (bottle P) very gently. Remove the paper test strip and note the length of the yellow-to-brown stain. A blank test shall be run using water in lieu of the arsenic solution; no stain shall form on the paper test strip.

4.5.1.2 Reagents D and E. Prepare, immediately prior to use, an aqueous solution containing 5 ppm nitrogen mustard. Rinse a test tube with the solution and carefully add 1 ml of the solution to the test tube. Within 3 minutes after preparing the agent solution, add 1 tablet from vial D to the test tube, then bring to a boil and allow to cool. Add 2 tablets from vial E to the solution in the test tube. Agitate the test tube gently and observe the solution against a white background for any color change. Any color change to blue or red, even though it fades rapidly, indicates that the tablets (reagents D and E) are satisfactory. A blank test shall be run using water in lieu of the mustard; no blue or red color shall appear in the water.

4.5.1.3 Reagent F. Prepare 3 buffered solutions of pH 4.5, 6.0, and 7.5 as determined with a pH meter. Immerse 3 reagent F pH test papers in each solution. The color of the test papers shall match the color for equivalent pH's shown in the instruction booklet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

4.5.1.4 Reagent G. Crush 3 halazone tablets from vial G and dissolve in 50 ml of water and an Erlenmeyer flask. Carefully add 2 ml of concentrated hydrochloric acid to the contents of the flask. Add 0.35 g of potassium iodide to the flask and dissolve. Titrate the contents of the flask with standardized 0.05 N sodium thiosulfate solution from a 5-ml microburet using starch as an indicator. Calculate available chlorine as follows:

$$\text{Milligrams of available chlorine per tablet} = 11.82\ AN$$

where: A = milliliters of sodium thiosulfate and
N = normality of sodium thiosulfate.

4.5.1.5 Reagents H, I, J, and K. Prepare, immediately prior to use, an aqueous solution containing 0.5 ppm GB and call this solution "A". Rinse bottle P and a test tube three times each with solution "A". Fill bottle P to the 25-ml mark with solution "A"; add 1 tablet from vial H and swirl to dissolve, observing the time required for solution. Half-fill the test tube with solution "A"; add 2 tablets from vial I and shake to dissolve, observing the time required for disintegration. Using scoop J, add 1 level measure of powder from vial J to the test tube and shake for 2 minutes. Pour the contents of the test tube into bottle P, cap, and shake for 2 minutes. Add the contents of one ampule from carton K to bottle P, cap, and shake an additional 2 minutes. Add a sufficient amount of additional solution "A" to bottle P to bring the contents of the bottle to just below the yellow band on the neck of the bottle, shake, and let stand for 5 minutes. The yellow color of the liquid in the neck of the bottle, adjacent to the yellow band, shall be equal to or darker in color than the yellow band when examined visually. A blank test shall be run using water in lieu of solution "A"; the solution shall be colorless or faintly yellow.

4.5.1.6 Active oxygen content of reagent J. Weigh approximately 0.2000 g of reagent J and transfer quantitatively to a 300 ml Erlenmeyer flask. Add approximately 80 ml of cold water, acidify with 20 ml of 1 N sulfuric acid, and titrate with 0.1 N potassium permanganate (note: acidification and titration should be carried out without interruption). Calculate the percent active oxygen as follows:

$$\text{Percent active oxygen} = \frac{0.8\ AN}{W}$$

where: A = milliliters of potassium permanganate,
N = normality of potassium permanganate, and
W = weight of sample in grams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

4.5.1.7 Reagent X. Prepare an aqueous solution containing 1.0 ppm of residual chlorine. Fill the plastic tube of vial X to the bottom of the yellow band with the residual chlorine solution. Add 1 tablet from vial X to the contents of the plastic tube and swirl until dissolved. The color of the solution shall match the color of the yellow band when examined visually (see 6.8).

4.5.2 Ampule volumetric contents. Check two ampules from the same kit for volumetric contents by emptying the ampules into a microburet at 68° ± 2°F.

4.5.3 Ampule leakage. The ampule shall be placed under water at 25° ± 3°F above ambient temperature for 3 minutes, with the sealed end facing upward. An emission of bubbles shall be considered as leakage.

4.5.4 Test vial for vial X color band. Prepare two solutions - solution #1 shall consist of 100 ml of 2 ppm (residual) chlorine solution; solution #2 shall consist of 0.0080 0 ± 0.0005 g of o-tolidine dihydrochloride made up to 100 ml with water. To 10 ml of solution #1 in the test vial X, add 10 ml of solution #2, shake, and visually compare the color of the band around the test vial with the color developed by the solutions.

4.5.5 Bottle P color band. Prepare a 0.0012 ± 0.0001 M potassium dichromate solution. Fill bottle P to the bottom of the neck with chloroform; add sufficient potassium dichromate solution to bring liquid level to the bottom of the yellow band. Visually compare the colored band of bottle P with the color of the dichromate solution.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. Each kit shall be packaged in accordance with method IC-2 of MIL-P-116. The box shall conform to class domestic, variety SW, grade 125, for a type 1 load of PPP-B-636. The inside dimensions of the box shall be 6-3/4 by 3-1/8 by 3-7/8 inches. Fiberboard pads shall be inserted in the box as needed to prevent motion of contents. Fiberboard pads shall conform to class domestic, variety SW, grade 125 of PPP-F-320. The inside dimensions of the barrier bag shall be 13-1/4 by 8-1/8 inches.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 Level A. Ten kits packaged as specified in 5.1 shall be packed in a fiberboard box conforming to type CF, class weather-resistant variety SW, grade W5c or W6c of PPP-B-636. The inside dimensions of the box shall be 19 by 7-1/4 by 8-7/8 inches. Fiberboard pads shall be inserted in the box as needed to prevent motion of contents. Fiberboard pads shall conform to grade W5c or W6c of PPP-F-320. The box shall be closed and reinforced in accordance with the closure and reinforcing requirements specified in the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. Ten kits packaged as specified in 5.1 shall be packed in the same manner as specified in 5.2.1, except the fiberboard box shall conform to type CF, class domestic, variety SW, grade 175 of PPP-B-636. The inside dimensions of the box shall be 19 by 7-1/4 by 8-7/8 inches. Fiberboard pads shall be inserted in the box as needed to prevent motion of contents. Fiberboard shall conform to type CF, class domestic, variety SW, grade 175 of PPP-F-320. The box shall be closed in accordance with the closure and reinforcing requirements specified in the appendix of PPP-B-636.

5.2.3 Commercial packing. The kits packaged as specified in 5.1, shall be commercially packed in accordance with Fed. Std. No. 356.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), unit packages shall be marked in accordance with MIL-STD-129. Shipping containers level A and B, shall be marked in accordance with Drawing D5-77-2348. When commercial packing is specified, shipping containers shall be marked in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Intended use. This kit is intended for use in determining the toxicological characteristics of drinking water.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and data of this specification.
(b) Level of packing required, (A, B or commercial).
(c) Preproduction sample size when required.

6.3 Arsenic test papers. Arsenic test papers are prepared by immersing the paper in alcoholic mercuric chloride solution for one hour. The alcoholic mercuric chloride solution consists of 5 grams mercuric chloride dissolved in 100 ml ethyl alcohol.

6.4 Polyethylene glycol 6000. Polyethylene glycol 6000 is a polymer of ethylene oxide with an average molecular weight of 6000 to 7500.

6.5 Ortho-dianisidine dihydrochloride. Ortho-dianisidine dihydrochloride is prepared as follows: dissolve 15 g technical grade ortho-dianisidine (melting point 137° to 140°C) in 150 ml of water, add 11.25 ml of concentrated hydrochloride acid, and heat the solution to boiling. Add 0.6 g of activated charcoal to the solution and boil for 8 minutes; add 0.75 ml of 32 percent aqueous sodium bisulfite solution, boil for 2 minutes, and filter through a Buchner funnel using acid-resistant filter

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

paper. Repeat the additions of charcoal and sodium bisulfite, the boilings, and filtration procedure three times. To the filtrate, add 30 ml of concentrated hydrochloric acid (containing 0.75 ml of 32 percent aqueous sodium bisulfite), cool to approximately room temperature, and filter off the precipitate (note: no more than 30 minutes shall elapse between the start of cooling the filtrate and the collecting of the precipitate). Wash the collected precipitate with 80 percent ethanol; follow with a wash of absolute alcohol. Vacuum dry the collected precipitate at 50°C.

6.6 Sodium hexametaphosphate. Because of the variations between batches of sodium hexametaphosphate, the requirements of 3.1.2.9 can best be met by using that sodium hexametaphosphate which gives a solution having a pH of no less than 7.7 when 0.39 g of sodium hexametaphosphate and 0.010 g of potassium carbonate are dissolved in 25 ml of water.

6.7 Acetone. The acetone used in the preparation of reagent K must be free of all oxidizable substances if the reagent is to perform satisfactorily when used as specified in 4.5.1.5. Satisfactory acetone is prepared as follows: Add 25 g potassium permanganate to 5 gallons acetone, shake well, and keep in a dark place for no less than 14 days. Shake once per day during this period. Decant the liquid or filter off most of the sediment. Distill the liquid using a 5-liter round-bottom flask heated by a variac-controlled electric heating mantle. Connect the distillation flask to a 15-inch Vigreaux distilling column with a Claisen distilling head and a water-cooled, 400-mm West condenser. Insert a thermometer in the side neck of the Claisen head and glass-plug the other opening; all connections must be glass and glass beads should be used in the distillation flask to prevent bumping. Rejecting the first 100 ml of distillate, collect the distillate boiling 55.5° to 56.5°C in a clean amber bottle; reject the last 200 ml of distillate. Test each batch of acetone distillate as follows: To 50 ml of water and 10 ml of acetone, add 2 ml of one percent aqueous orthotolidine dihydrochloride (white crystalline) solution and 1 ml of 0.5M potassium phophate, dibasic. Mix thoroughly and add 5 ml of one percent Perdox (low ratio). Mix thoroughly and add 10 ml xylene. Mix thoroughly by inverting approximately 20 times. Allow the xylene layer to separate and read an 8 ml aliquot on a photoelectric colorimeter at 420 millimicrons using xylene as a blank to zero the instrument. The xylene layer should be colorless visually and should have a percent transmission of no more than 10 on the colorimeter. If a colorimeter is not available, the xylene layer should be compared with an equal volume of a color standard (0.000048M potassium dichromate solution).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

6.8 Residual chlorine. For determining 1 ppm residual chlorine, the result obtained (4.5.1.7) should be in agreement with that determined by using orthotolidine solution prepared and applied in accordance with Standard Methods of Water Analysis, American Public Health Association, 8th edition, 1936.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6665-A355

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

APPENDIX

INSTRUCTION BOOKLET, WATER TESTING KIT,

CHEMICAL AGENTS, AN-M2

10. SCOPE

10.1 This appendix covers the instructions to be included with each AN-M2 Water Testing Kit, Chemical Agents.

20. APPLICABLE DOCUMENTS

20.1 The following specification of the issue in effect on date of invitation for bids form a part of this appendix:

FEDERAL

UU-P-121 - Paper, Bond and Writing, White and Colored.

30. REQUIREMENTS

30.1 Paper. The paper used for preparation of the instruction cards shall be white and shall conform to Specification UU-P-121.

30.2 Printing. Printing shall be accomplished with black printer's ink and shall be completely legible.

30.3 Size. The instruction cards shall measure not more than 2-3/4 by 6 inches with each sheet laminated and heat sealed over its entire surface between sheets of plastic. Plastic sheet shall be composition B, type I, class 2, 0.005 nominal thickness, clear, matte finish conforming to L-P-5350. All sheets shall have a 1/8 inch diameter hole in one end through which twine shall be tied to keep cards together. Twine shall be nylon no. 15 treated, breaking strength 115 pounds, 4 $\pm$ 1/2 inches long.

30.4 pH color chart. The booklet shall contain a full page nitrazine pH color chart. The range shall be from 4.5 to 7.5 in 0.5 pH increments. This chart shall appear as the page immediately following the page containing the instructions for the pH test (paragraph 3).

30.5 Information. The booklet shall contain the following information, stated exactly as herein specified:

DIRECTIONS FOR USE OF THE

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

WATER TESTING KIT, CHEMICAL AGENTS, AN-M2 1/

1/ This title shall appear on both front and back pages.

I. PURPOSE

1. This field kit for water testing and screening is designed as a reconnaissance kit. Its purpose is to screen out raw water sources so contaminated with chemical agents that they cannot be rendered potable by customary field treatment methods, such as chlorination in the Lyster Bag.

2. Negative tests indicate that the water may be used sparingly by troops for a period not exceeding one week after treatment by customary field methods.

3. If any of the tests are positive, the water should not be used until a more complete analysis can be made.

4. The main purpose of the kit is to detect contamination by chemical agents in raw water. It is not designated for use in the control of treated water. Chemical reactions during water treatment invalidate the interpretations.

II. TEST PROCEDURES

A. General Directions.

1. Read directions thoroughly. All steps must be performed in the exact order given.

2. Obtain a water sample in a canteen cup without excessive disturbance of the water source. Sample should be taken as nearly as possible at the point and depth from which the water is to be taken. This water will be used for all tests except the chlorine demand test (Para II, E. 1.).

3. After using any of the tablets or reagents, be sure to immediately close the vial with the matching colored cap.

4. Start the arsenic test (Para B.). While the arsenic test is developing, perform the other tests.

B. Arsenic test.

1. Procedure.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

a. Fill the bottle marked P to the upper (25 ml) mark with water from the canteen cup.

b. Place 2 tablets from the light green capped vial A into the bottle. Swirl to dissolve.

c. Take 1 test strip from the yellow capped vial B and carefully insert it into the tube L, bending the strip near the top so it will remain upright in the tube. Touch only the top end of the strip. Keep test strip dry.

d. When the tablets from the light green capped vial A have dissolved, add 5 pellets from the brown capped vial C to the contents of bottle P, and promptly fit the tube assembly into the bottle (see fig. 1).

e. Let react for 20 minutes. If the solution is cold, warm the bottle in your hands.

f. Remove the test strip from the tube L and note the length of the yellow to brown stain. A stain 1/4 inch or more indicates a positive test; a stain less than 1/4 inch indicates a negative test.

C. Mustard Test (Including Nitrogen Mustards and Cyanogen Chloride).

1. Procedure.

a. Rinse a test tube with the suspected water, and carefully fill the tube to about 1/2 inch depth with the suspected water.

b. Add 1 tablet from the blue capped vial D and agitate for 3 minutes. Bring the solution to a boil with a match, lighter, or other source of heat. (The test carried out without the flame is less sensitive but will detect amounts of nitrogen mustards likely to cause serious casualties.) Allow the solution to stand for 5 minutes. During cold weather keep the tube warm in hand or inside pocket for an additional 5 minutes. A yellow color in the solution is positive for cyanogen chloride.

c. Add 2 tablets from the red capped vial E to the test tube. Agitate the test tube gently, observing against a white background for any change of color in the liquid (or in the curd) from white to red or blue. Any visible color change, even if it fades rapidly, indicates a positive test for mustard or nitrogen mustards. A yellow color indicates cyanogen chloride. A white color indicates a negative test.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

D. pH test.

1. Procedure. Dip 1 strip of test paper from the black capped vial F into the suspected water. Remove and compare the resulting color of the test paper with the color chart on the opposite page. pH of less than 6 indicates possible contamination. (Print full page of nitrazine pH color chart, range from pH 4.5 to pH 7.5 inclusive in intervals of 0.5 pH units on opposite page.)

E. Chlorine demand test.

1. Procedure.

a. Fill a canteen with the suspected water to within 1 inch of the top, add 3 tablets from the white capped vial G, screw on the cap, and shake vigorously for 5 minutes. Allow the solution to stand for 5 minutes.

b. Fill the test vial X to the bottom of the yellow band with water from the canteen. Add 1 tablet from the black capped vial S, swirl until it is completely dissolved and compare the color of the solution with that of the yellow band on the tube.

c. A positive test is indicated by no color or color lighter than the yellow band. A negative test is indicated by an orange color or a color darker than the yellow band.

F. G Agents.

1. Procedure. (DO NOT PERFORM THIS TEST IN DIRECT SUNLIGHT).

a. Rinse bottle P with the suspected water 3 times, fill the bottle to the second mark with the water, and add 1 tablet from dark green capped vial H. Swirl the solution to dissolve the tablet. If a yellow or green color appears, discard the solution and test as prescribed in para g. If no yellow or green color is formed, continue as follows:

b. Rinse one test tube 3 times with suspected water from the canteen cup, and fill the test tube 1/2 full with the suspected water. Add two tablets from the orange capped vial I to the test tube and shake to dissolve.

c. With scoop J add 1 level measure of powder from gray capped vial J to the test tube and shake for 2 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

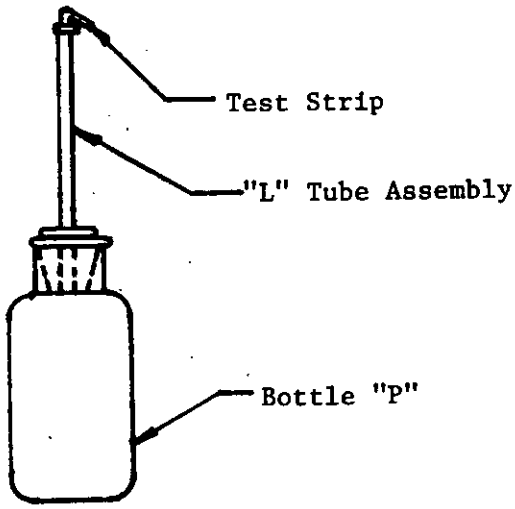


Figure 1 - "L" Tube Assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

d. Pour the contents of the test tube into the bottle P, place the plastic cap on the bottle and shake well for 2 minutes.

e. Break the tip of one of the ampules in carton K and add the contents to the bottle. Replace the plastic cap and shake for an additional 2 minutes. Add enough water from the canteen cup to bring the top level of the solution into the neck of the bottle just below the yellow band and let stand for 5 minutes.

f. A yellow color in the upper layer equal to or darker than the yellow band on the bottle is a positive test for G agents and for certain toxic insecticides. A yellow color in the lower water layer indicates very dangerous concentrations of G agents. With time, the yellow band on the bottle may fade, in this case, a comparison may be made with the yellow band on the test vial used for the chlorine demand test.

g. Rinse the bottle P and fill to the second mark on the bottle. Add 2 tablets from the orange capped vial I and shake to dissolve. Add 1 level scoop of powder from the gray capped vial J followed immediately by one tablet from the dark green capped vial H. Place the plastic cap on the bottle and shake well for 2 minutes. Continue as specified in paragraph e and f.

h. In the absence of a highly positive chlorine demand test, no color or a color lighter than the yellow band indicates a negative test for G agents.

G. Taste and odor.

1. Procedure.

a. If the arsenic test B, mustard test C, chlorine demand test E, and G-agents test F, are negative and if the pH is 6 or above, carefully smell and taste a sample of the suspected water.

b. A positive test is indicated by:

(1) A lacrimatory or chlorine odor.
(2) A biting and/or peppery chlorine taste.
(3) Any taste or odor of a known war gas.

c. Absence of all tastes and odors will indicate a negative result, but not necessarily a safe water. A negative test is also indicated by the presence of only those odors and/or tastes normally characteristic of natural waters.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.

H. Interpretations.

Table 1. Interpretations of tests

Test	Contaminated Water	Non-Contaminated Water
Arsenic test, B	positive	negative
Mustard test, C	positive	negative
pH test, D	pH below 6	pH above 6
Chlorine demand test, E	positive	negative
G-agents test, F	positive	negative
Taste and odor test, G	positive	negative

If any of the above tests are positive or the pH is below 6, the water will be considered to be contaminated; if the tests are all negative and the pH is above 6, the water will be considered suitable for one week after bacterial disinfection by usual methods.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-51114A(EA) Water Testing Kit, Chemical Agents, AN-M2

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:50Z
Check the source to verify that this is the current version before use.